

PHYSICAL DESIGN ASSIGNMENT

1. What are the techniques to avoid dynamic power ?

- Dynamic power = $\alpha \cdot C_{load} \cdot V_{dd}^2 \cdot f$
- Sequential Circuits which receives clock even when it is idle can be made as a gated clock to reduce (α – switching activity) power consumption, for that a and gate with two inputs clock and enable is used where enable = 0 means it won't allow the clock signal to pass the circuit and enable = 1 means only in that time span window it allows clock to pass to sequential circuit.
- $C_L \approx C_{interconnects} + C_{gate}$ where this interconnects capacitance can be reduced by planning the floorplanning stage very well and this $C_{gate} = C_{ox} \cdot W \cdot L$ where length is technology it can't be changed and changing in width will affect the RC value which leads to decrease in dynamic power but increase in delay is a trade off and this is suitable only for the non-critical paths gates.
- Reducing V_{dd} is one of the most effective ways to reduce dynamic power because of the square dependence. However, lowering V_{dd} beyond a certain point can cause timing violations (the circuit may fail to meet speed requirements) because transistors operate slower at lower voltages.
- Reducing the clock frequency can also reduce power consumption linearly. However, it directly affects the throughput and speed of the system, so it's usually done only during periods of low workload or idle states.

2. Why double via insertion ?

- Double via insertion is used in physical design to improve the **reliability** and **manufacturability** of integrated circuits. A via is a small conductive connection between two metal layers in a chip. The reason why double vias are important:

□ **Improves Reliability:**

- Single vias can fail due to manufacturing defects or stress during chip operation.
- **Double vias provide a backup path**, ensuring connectivity even if one via fails.

□ **Reduces Resistance Variations:**

- Vias have some resistance, and defects in a single via can increase resistance significantly.

- Using double vias **reduces resistance variations, improving signal integrity** and circuit performance.

❑ **Enhances Yield in Manufacturing:**

- Chip manufacturing processes can introduce defects due to alignment errors, process variations, or contamination.
- Double vias **increase the probability of a successful connection, improving overall yield.**

❑ **Critical for Advanced Nodes:**

- At smaller technology nodes (e.g., 7nm, 5nm), vias become smaller and more prone to defects.
- Double vias are a common strategy to address these challenges.

❑ **Minimizes Electromigration Risks:**

- Electromigration, caused by high current densities, can degrade single vias over time.
- **Double vias distribute current more evenly, reducing the risk of failure.**

Double via insertion is a simple yet effective technique to enhance the reliability, performance, and yield of a chip during and after manufacturing. It's particularly important in advanced technology nodes and high-performance designs.

3. Mosfet and FinFET difference?

Aspect	MOSFET	FinFET
Structure	Has a planar structure with a flat channel.	Has a 3D fin-like structure for the channel.
Gate Control	Single gate controls the channel.	Gate surrounds the channel on three sides, providing better control.
Leakage Current	Higher leakage current, especially at smaller nodes.	Lower leakage current due to improved gate control.
Performance	Slower switching speed at advanced nodes.	Faster switching and better performance.
Scaling	Difficult to scale below 20nm due to short-channel effects .	Scales well below 20nm, suitable for advanced nodes.

Also Mosfet stands for Metal Oxide Semiconductor and Field Effect Transistor while FinFET stands for Fin Field Effect Transistor.

4. What is CMP?

In Physical Design (PD) in VLSI, **CMP** stands for **Chemical Mechanical Planarization** or **Chemical Mechanical Polishing**.

- **Purpose:** CMP is a process used in semiconductor manufacturing to achieve a flat and uniform surface across a wafer.
- **Key Role:** It is essential for:
 - Removing excess material after deposition.
 - Flattening layers to ensure proper lithography and interconnect fabrication.
- **Process:** Combines chemical etching and mechanical polishing to smooth and planarize the wafer surface.

Importance of CMP in PD:

1. Ensures uniformity for subsequent layers in multilayer designs.
2. Improves the quality of interconnect layers by reducing defects.
3. Enables the scaling of devices by supporting advanced process nodes.

CMP is a critical step in achieving high-yield and reliable VLSI chips.

5. Why should we define NDRs during placement ?

Defining **NDRs (Non-Default Rules)** during the placement step in physical design is important because it helps improve the quality, performance, and reliability of the chip. NDRs are used to set special rules for wire widths, spacings, and vias for critical nets, and here's why they matter:

1. **Protect_Important_Nets:**
Some nets, like clock or power signals, are very sensitive and need extra care. Using NDRs makes sure these nets get wider wires and more spacing, so they don't face problems like high resistance or noise.
2. **Improve_Signal_Quality:**
Wider wires and proper spacing reduce issues like crosstalk (when one signal interferes with another). This is especially helpful for high-speed or critical signals.
3. **Avoid_Congestion:**
If you apply NDRs during placement, it ensures that the layout won't get too crowded in certain areas, which makes routing easier later on.
4. **Better_Timing:**
By setting rules for critical paths early, you can improve timing because wider wires have less resistance, and proper spacing reduces delays.

5. **Save_Time_and_Effort:**

If NDRs aren't defined during placement, you might face issues during routing, like violations or delays, which means extra work to fix them. Doing it early avoids this hassle.

Simple Example:

If I think a chip design like a road map. Some roads (critical nets) will have more traffic (signals), so we need to make those roads wider and give them more space to avoid jams. Setting these rules early helps the flow stay smooth without needing to redo things later.

By defining NDRs during placement, we're making sure the design process is smoother and the final chip works better. It's like planning ahead so you don't run into problems down the line.

6. What are Electro Migration (EM) fixing methods ?

Electromigration (EM) occurs when high current density causes metal atoms in wires to shift, leading to reliability issues. Here are simple methods to fix EM during physical design:

1. **Widening Wires:** Increases wire width to reduce current density.
2. **Adding Redundant Vias:** Uses extra vias to share the current load.
3. **Using Parallel Wires:** Splits high currents across multiple wires.
4. **Thicker Wires:** Thicker metal layers handle more current.
5. **Shorter Wire Lengths:** Splits long wires and uses buffers.
6. **Current Redistribution:** Balances current flow by rerouting.
7. **Improving Power Network (PDN):** Adds power rails and vias for high-current paths.
8. **Following EM Design Rules:** Adheres to foundry guidelines for current limits.
9. **Spacing Adjustment:** Increases spacing for better thermal management.
10. **Top Metal Layers:** Routes high-current signals on thicker top layers.
11. **EDA Tools for EM Fixes:** Uses tools to analyze and fix EM issues automatically.

These are the techniques which ensures the design is robust and reliable.

7. What is MCMM ?

MCMM stands for **Multi-Corner Multi-Mode**, a critical concept in the physical design of integrated circuits (ICs). It refers to the process of ensuring a chip performs reliably across various operating conditions (corners) and functional scenarios (modes).

Breaking Down MCMM:

1. Multi-Corner (MC):

- A "corner" represents a specific combination of **process, voltage, and temperature** (PVT) variations.
- For example:
 - **Process corners:** Slow-Slow (SS), Fast-Fast (FF), Typical-Typical (TT).
 - **Voltage corners:** Minimum voltage, nominal voltage, maximum voltage.
 - **Temperature corners:** Low temperature, high temperature.
- Each corner simulates the chip under different conditions to ensure it meets timing, power, and functionality requirements in real-world scenarios.

2. Multi-Mode (MM):

- A "mode" represents different operational states of the chip, such as:
 - Functional mode: Normal chip operation.
 - Test mode: Chip operation during manufacturing tests.
 - Low-power mode: Reduced performance for saving power.
- Different modes have unique constraints and timing requirements, which must be validated.

Importance of MCMM:

- Modern ICs are designed for **performance, power, and reliability** across various use cases and environments.
- MCMM analysis ensures that:
 - Timing constraints (setup and hold times) are met across all corners and modes.

- The chip operates correctly in worst-case scenarios (e.g., high temperature and low voltage) and best-case scenarios (e.g., low temperature and high voltage).

Example:

Consider a chip designed for a smartphone:

- **Corners:** It must function correctly in extreme conditions like 85°C (hot) or -40°C (cold), with voltages varying from 0.9V to 1.2V.
- **Modes:** It must support functional operation (normal usage), test mode (manufacturing), and low-power mode (when idle).

MCMM analysis ensures a robust and reliable chip design by validating performance across all critical scenarios. It is an essential step in achieving a high-quality IC that functions correctly in real-world conditions.